

29/03/2026

sparse representation;
context capture
(semantic meaning);



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vectors should
size be fixed.

One hot encoding
Word to class frequency.

The cat sit on the mat.
The dog sit on the mat.

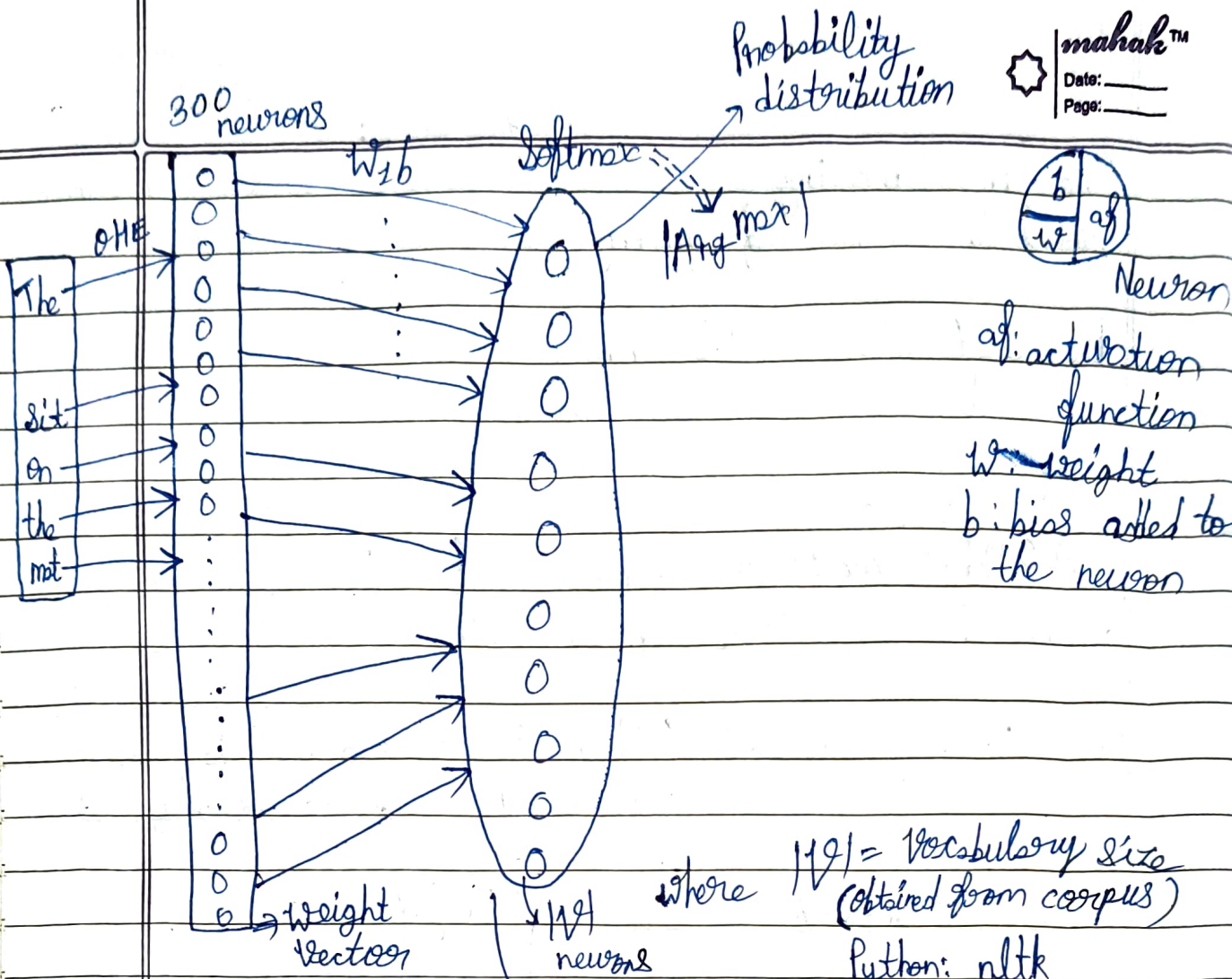
Word2Vec: word surrounded by similar word
must have similar representation

→ Semi-Supervised (Neural Network + Data (Corpus) without any label)

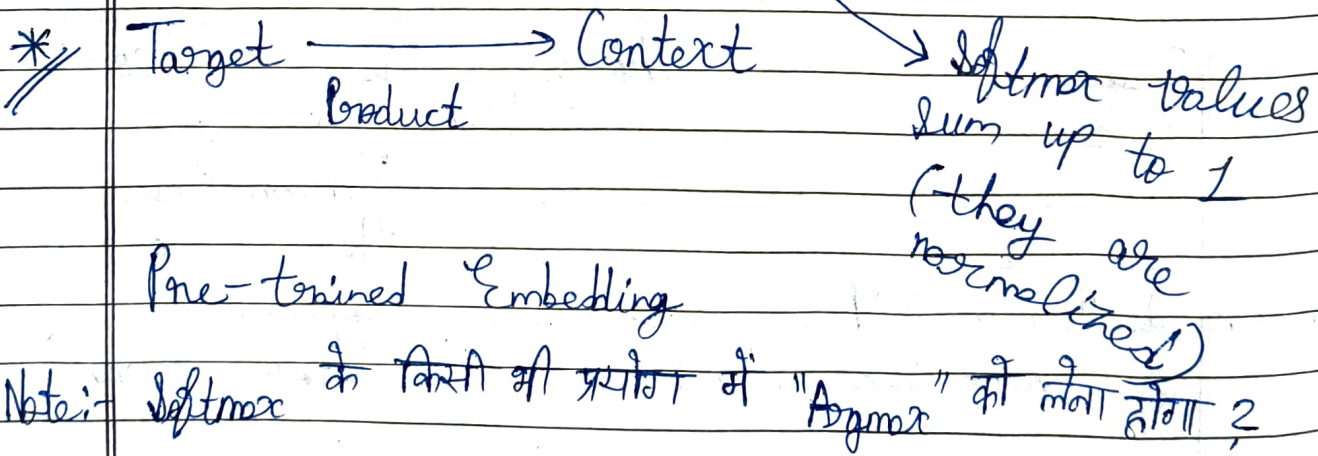
* Context	Target	
The ___ sit on the mat.	cat	} CBOW: (continuous Bag of words)
The ___ sit on the mat.	dog	
I am ___ to Lucknow.	going	

Additionally,
there is one more
approach
(Skip-gram)

Note:- 2-layer NN is used in
the Word2Vec schema.



Skip-gram



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Glove: Global Vectors for Word Representation

word-word co-occurrence matrix

- I enjoy flying
- I like NLP
- I like deep learning

	I	like	enjoy	deep	learning	NLP	flying
I	0	2	1	0	0	0	0
like	2	0	0	1	0	1	0
enjoy	1	0	0	0	0	0	1
deep	0	1	0	0	1	0	0
learning	0	0	0	1	0	0	1
NLP	0	1	0	0	0	0	1
flying	0	0	1	0	0	0	1
	0	0	0	0	1	1	1

$$X = X^T$$

(a symmetric matrix)

X_{ij} = Number of times word j appearing with word i

$$x_i = \sum_{k=1}^{|V|} x_{ik}$$

$$P_{ij} = P(w_j | w_i) = \frac{X_{ij}}{\sum_{i=1}^{|V|} \pi_i}$$

$$\therefore P(\text{like} | I) = \frac{\text{count}(I, \text{like})}{\text{count}(I)} = \frac{2}{3}$$

Word2Vec:-

- Only local view of data.
- Local window may capture word similarity
- This information is stored in updated weights.

GLoVe:

- word co-occurrence matrix
- capture global information

Target word: ice, steam

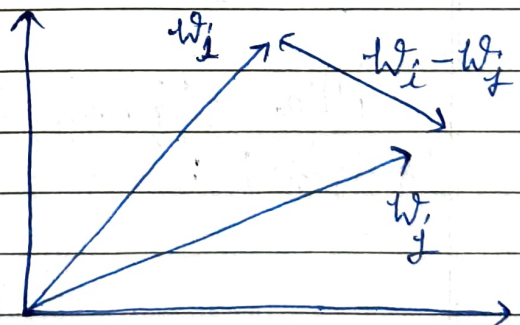
- Context words from 6 billion tokens
- Solid, gas, water, fashion
- Solid & gas should be related with ice & steam
- Water and fashion are unrelated.

Probability Ratio	K=Solid	K=gas	K=water	K=fusion
$P(K/\text{Ice})$	1.9×10^{-4}	6.6×10^{-5}	3×10^{-3}	1.7×10^{-5}
$P(K/\text{steam})$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$\frac{P(K/\text{ice})}{P(K/\text{steam})}$	8.9	8.5×10^{-2}	1.36	0.96

$P(w)?$

$$F(w_i - w_j, w_k) = \frac{P_{ik}}{P_{jk}} \quad (\text{Eqn. (2) in Gilolo paper})$$

(CS224 Stanford University
Prof. Christopher Manning's
NLP Lectures YouTube)



Gilolo: Ratio of probabilities carry better information to able to distinguish solid and gas with water.



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$$F(w_i, w_j, \hat{w}_k) = \frac{P_{ik}}{P_{jk}}$$

~~word~~
vectors

(Eqn (1))
in paper

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Global Vectors (GloVe)

2014 published

Count-based approach

- One-Hot Encoding
- Word to frequency based representation
- TF-IDF based representation

Prediction-based approach

- Word2Vec (2013)
- Semi-supervised (2 large NNs)
- Context $\leftrightarrow$ Target
 - ↳ CBOW
 - ↳ Skipgram

GloVe = Counting-based approach + Prediction-based approach

- word-word co-occurrence matrix.
- Ratio of co-occurrence probabilities can encode word meaning.

Co-Occurrence Matrix (X)

Eg:-

	I	love	NLP
I	_____	_____	_____
love	_____	_____	_____
NLP	_____	_____	_____

	$x = \text{solid}$	$x = \text{gas}$	$x = \text{water}$	$x = \text{fashion}$
$P(x/\text{ice})$	large	small	large	small
$P(x/\text{steam})$	small	large	large	small
$\frac{P(x/\text{ice})}{P(x/\text{steam})}$	large	small	~ 1	~ 1

For the two words i and j , assume their corresponding vector is w_i and w_j .

$$F(w_i, w_j, \tilde{w}_k) = \frac{P_{ik}}{P_{jk}} \quad (w_i - w_j \text{ representation})$$

means, in 2-D space.

$$F(w_i - w_j, w_k) = \frac{P_{ik}}{P_{jk}}$$

$\propto \frac{1 \times 300}{\sqrt{300 \times 1}} \frac{1 \times 300}{\sqrt{300 \times 1}}$

$$\frac{(w_i - w_j)^T \cdot w_k}{(1 \times 300) (300 \times 1)}$$

(log-probability Here)

$$w_i^T \cdot w_j = \log P(j/i)$$

Similarity How likely is j to occur in context of i .
 between i and j



$$w_i^T w_j = \log \left(\frac{x_{ij}}{x_i} \right) = \log x_{ij} - \log x_i \quad (1)$$

$$\text{Similarly, } w_j^T w_i = \log \left(\frac{x_{ij}}{x_j} \right) = \log x_{ij} - \log x_j \quad (2)$$

Add (1) and (2),

$$2 w_i^T w_j = 2 \log x_{ij} - \log x_i - \log x_j$$

$$w_i^T w_j = \frac{\log x_{ij}}{2} - \frac{\log x_i}{2} - \frac{\log x_j}{2}$$

Bias with
 w_i

Bias with w_j

b_i = bias associated with word i (w_i) bias associated with word j (w_j) = b_j

$$\boxed{w_i^T w_j + b_i + b_j = \log x_{ij}} \quad (\text{Eqn. (7) in GloVe paper})$$

Loss function: $\min \sum_{i,j} (w_i^T w_j + b_i + b_j - \log x_{ij})^2$

$[w_i, w_j, b_i, b_j] \checkmark$

(continued... after 13 march notes)

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Problem:

———— Gives equal weightage to every co-occurrence

Ideally, more and very frequent co-occurrences should have less weightage

Modification:-

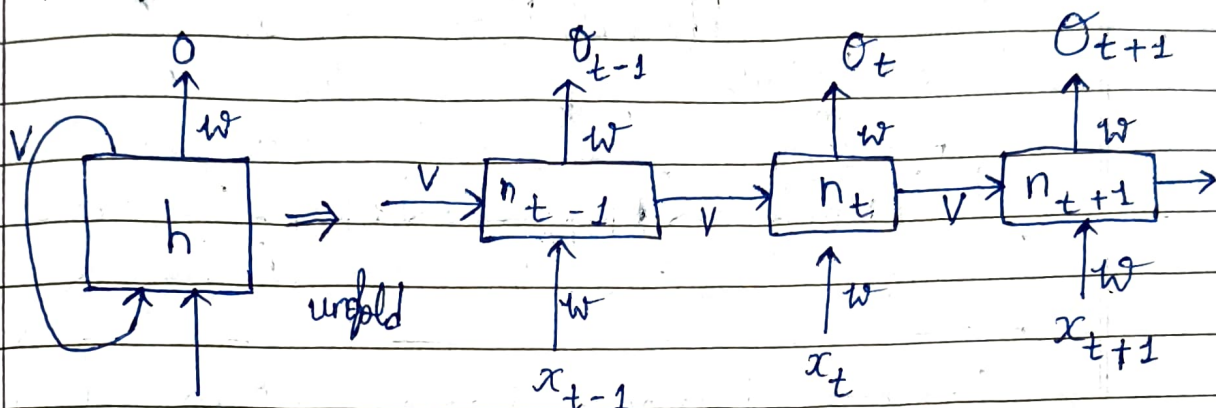
$$\text{Min.}_{w_i, w_j, b_i, b_j} \sum_{ij} f(x_{ij}) (w_i^T w_j + b_i + b_j - \log x_{ij})^2$$

(Eqn. (8) in GloVe Paper).



23/03/2026

Recurrent Neural Network



Feed-Forward
Neural Network

Sequential data, time series data

← FFNN

RNN

1. No loops	Loop through time
2. No memory of past i/p's	Maintain memory via hidden state.
3. Input are independent.	Inputs are sequential dependent.
4. Static layers, no time dimension.	Unfold across timesteps.
5. Standard backpropagation.	Backpropagation through Time (BPTT)

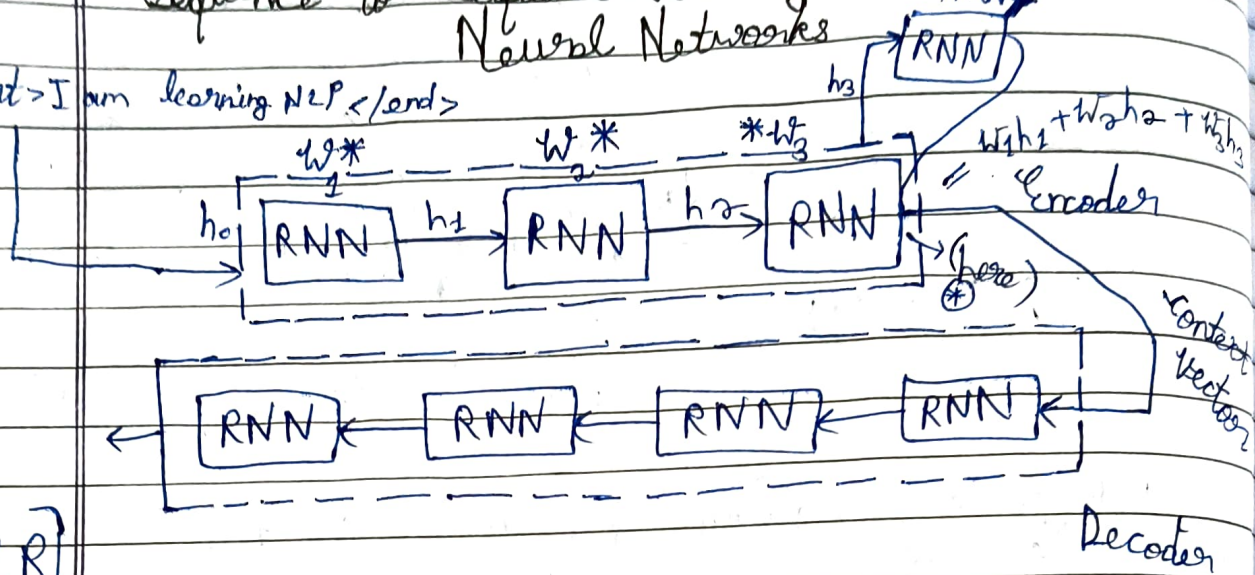
LSTM	1997
GRU	2014
Bi-LSTM	
Bi-GRU	

→ Long Short-Term Memory
Gate recurrent unit

RNN

Sequence to Sequence Learning with Neural Networks

<Start> I am learning NLP </end>



ICLR 2015 Paper] This "encoder-decoder" architecture is used in many aspects such as image captioning, video captioning, chatbot, Q/A models, text-to-image generation, voice-to-text, etc.

Neural Machine Translation by jointly learning to align and translate.

Bahdanau Attention

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BPTT

(Backpropagation Through Time)

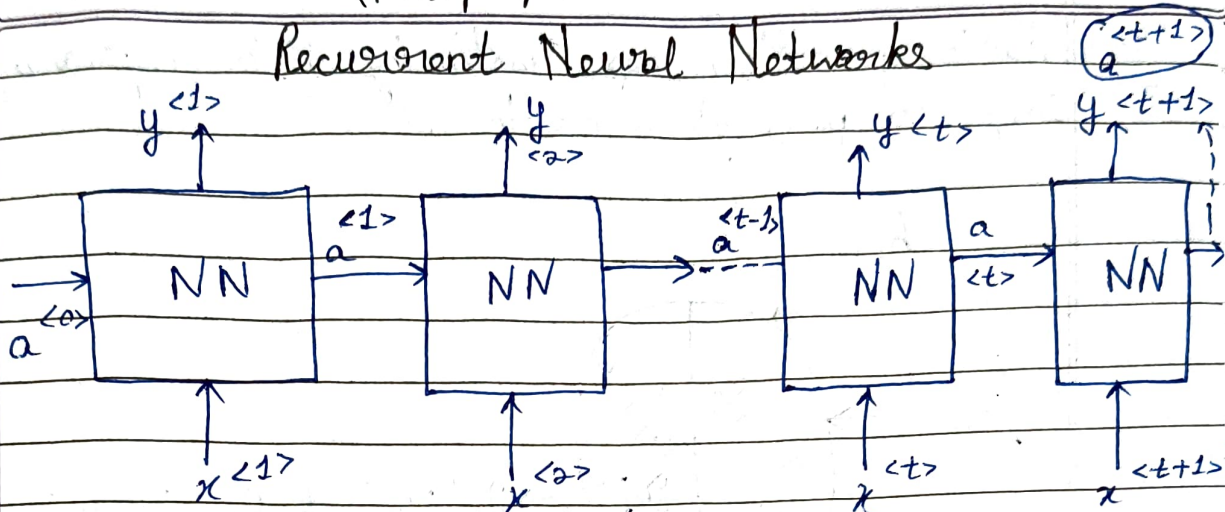


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Recurrent Neural Networks



For each time step t , the activation $a^{<t>}$ and the output $y^{<t>}$ are computed as follows:

$$a^{<t>} = g_1 \left(w_{aa} a^{<t-1>} + w_{ax} x^{<t>} + b_a \right)$$

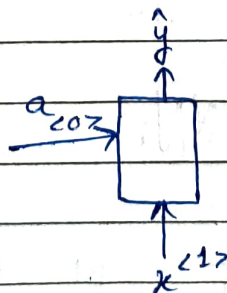
$$y^{<t>} = g_2 \left(w_{ya} a^{<t>} + b_y \right)$$

*

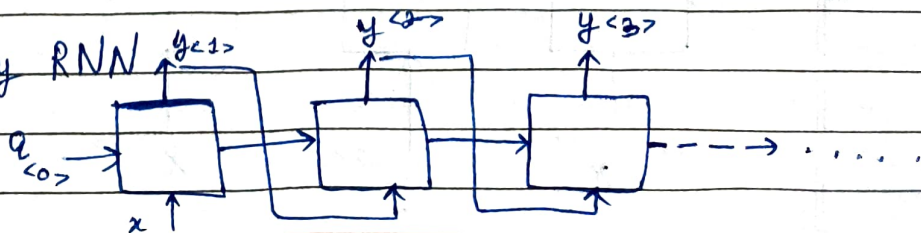
$$w_{ax}, w_{aa}, w_{ya}, b_a, b_y$$

are learnable parameters that are stored temporally

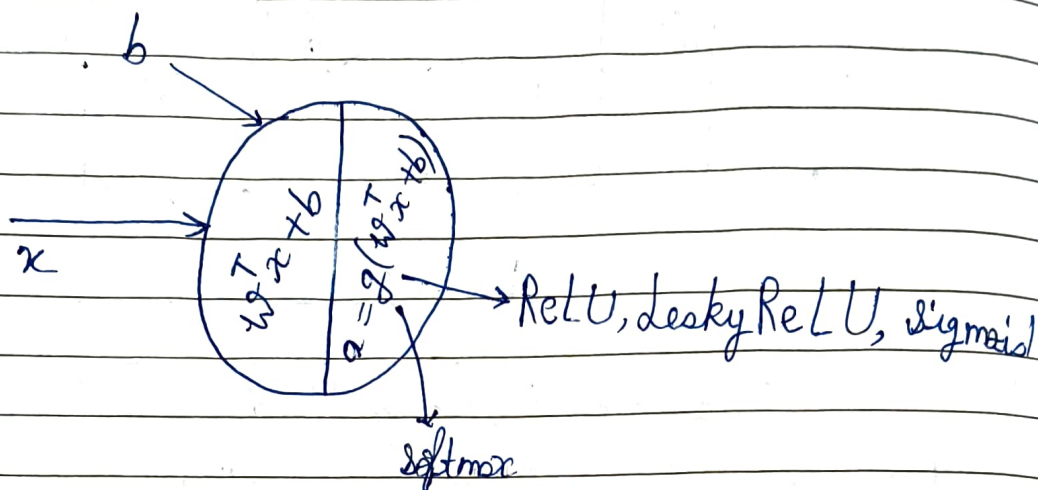
1.) One to One RNN



2.) One to Many RNN



Uses: Music generation, image captioning, video captioning, image to paragraph generation, etc.



Magnitude (Loss) = $|y - \hat{y}|$
(actual) (predicted)

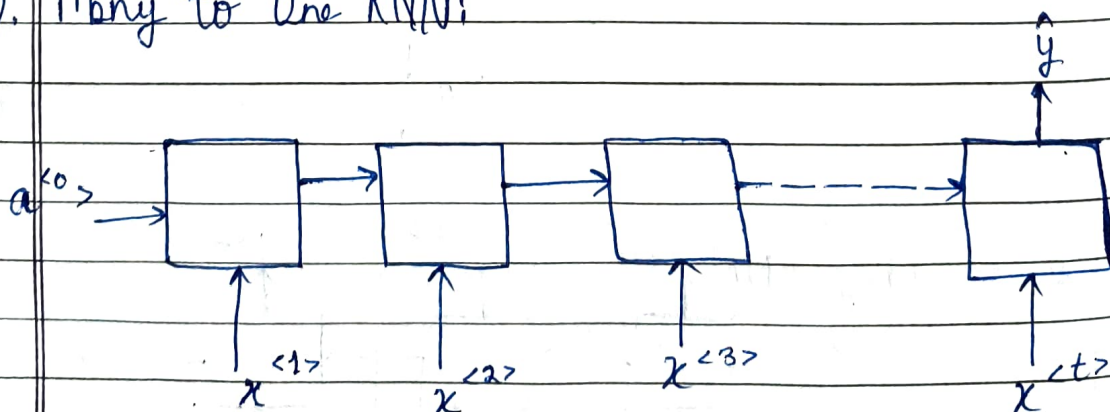
"Tuning of w and b "

↳ Adam / Stochastic Gradient Descent, etc. methods

→ $w = w - \alpha \frac{dw}{dL}$

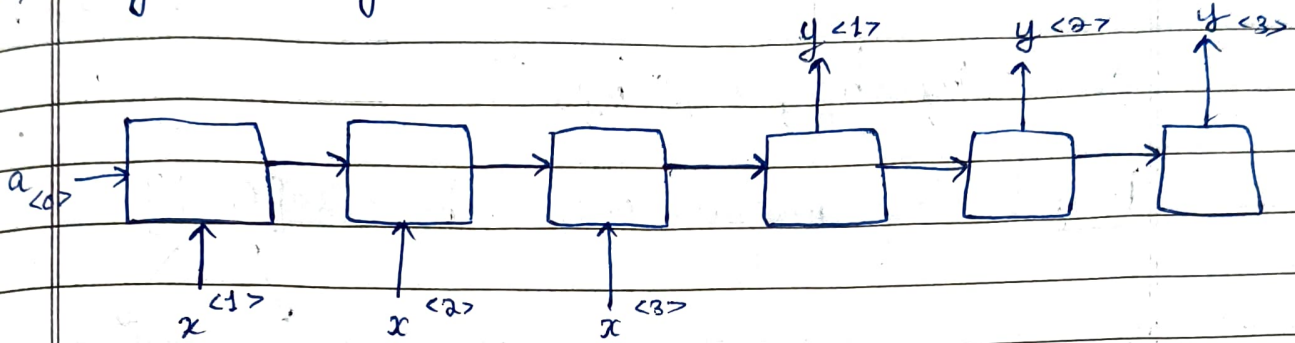
→ $b = b - \alpha \frac{db}{dL}$

3.) Many to One RNN:



Uses: Sentiment Analysis, Spam detection, Text classification,
 Forecasting Tasks,

4.) Many to Many



Uses: Machine Translation, Language Processing (Voice assistants like Siri, Alexa, Cortana, Google Voice, etc.)

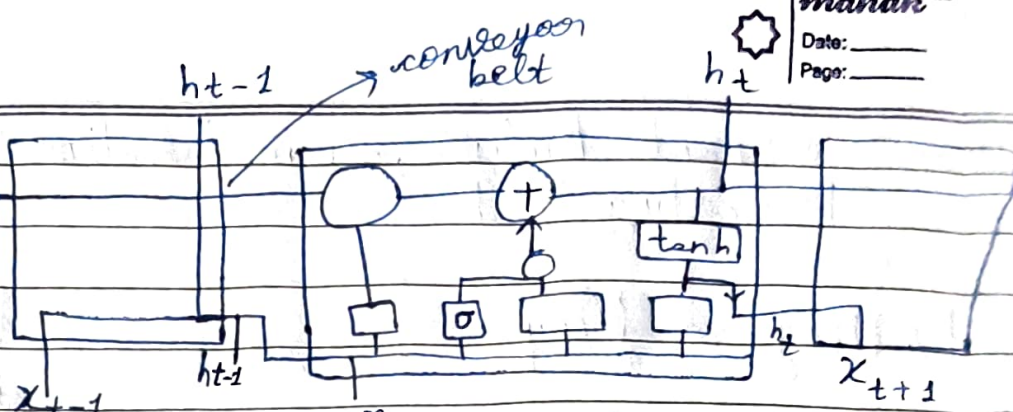
One confirmed NUMERICAL for BPTT
 in EndSem!!

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STM



$$f_t = \sigma(w_f \cdot [h_{t-1}, x_t] + b_f)$$

Forget Gate generates output between $[0, 1]$

Input Gate:-

Decides what new info. we are going to store in the cell state. sigmoid decides which values we have to update.

discard info | keep all info

$$i_t = \sigma[w_i * [h_{t-1}, x_t] + b_i]$$

$$\tilde{C}_t = \tanh h [w_c * [h_{t-1}, x_t] + b_c]$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

$$o_t = \sigma(w_o (h_{t-1}, x_t) + b_o)$$

$$h_t = o_t * \tanh h (C_t)$$